

# Md Abdullah Al Mazid

Miami, Florida | mazid.rafee@gmail.com | (786) 257-7354 | mazid-rafee.github.io

Dear Hiring Committee,

I am a PhD candidate in Computer Science at Florida International University, conducting research in the High Performance Database Research Center under Prof. Naphtali Rishe, with expected graduation in Spring 2027. I am seeking postdoctoral, research, and faculty opportunities where I can contribute to research at the intersection of multimodal machine learning, computer vision, scientific machine learning, and environmental and Earth-observation applications.

My research focuses on developing machine-learning methods that integrate heterogeneous data sources, physical knowledge, and interpretable representations. My dissertation research, **MSCloudr**, investigates conditional diffusion-bridge methods for reconstructing cloud-obscured Sentinel-2 imagery using Sentinel-1 SAR as complementary structural information. This work explores mean-reverting diffusion dynamics, reliability-aware multimodal conditioning, cross-modal alignment, and risk-aware trajectory design for robust optical-SAR fusion.

My related research includes **MSCloudCAM**, a multispectral cloud-segmentation framework combining transformer representations, multi-scale contextual modeling, and cross-attention across Sentinel-2 and Landsat-8 imagery. I also developed **pKANrtm**, published in *Remote Sensing*, a physics-aware multi-fidelity learning framework that combines 6S simulations with high-fidelity libRadtran modeling for atmospheric correction. The underlying methodology is also the subject of FIU Invention Disclosure D2026-0057. My broader research experience includes spatiotemporal environmental forecasting, interpretable visual models, prototype-based representation learning, and environmental data analysis.

In addition to research, I bring experience in teaching, competitive programming, and production software development. At FIU, I have supported **CIS 5027 - Computer Systems Fundamentals** and **COP 5725 - Principles of Database Management Systems** through instructional assistance, office hours, student discussions, assessment activities, and instructor-directed course development and review. During my undergraduate studies, I was actively involved in competitive programming, including participation in the **ICPC Dhaka Regional Final** and the **National Collegiate Programming Contest**. Competitive programming strengthened my algorithmic thinking, problem decomposition, debugging, and ability to develop efficient solutions under constraints-skills that later became valuable both in production software engineering and in designing and implementing research methods. As an educator, I would encourage interested students to develop these skills through programming contests, structured problem-solving practice, mentoring, and collaborative programming sessions, and I would be interested in helping organize and support competitive-programming activities within the department.

Before beginning my PhD, I spent more than three years in software engineering, including as a Senior Software Engineer at Enosis Solutions, where I worked on a large cloud-based SaaS platform, database development and optimization, software modernization, and technical leadership. This combination of research, teaching, competitive programming, and industry experience allows me to connect fundamental computer-science concepts with both research problems and production-scale software development.

My future research agenda centers on reliable multimodal learning, interpretable AI, and scientific machine learning, with particular interest in Earth observation, environmental sensing, and other domains where machine-learning systems must operate with incomplete, heterogeneous, or physically constrained data. I am particularly interested in opportunities that combine foundational methodological research with meaningful scientific or real-world applications.

I would welcome the opportunity to discuss how my research, teaching, and software-engineering background could contribute to your organization. Thank you for your consideration.

Sincerely,

Md Abdullah Al Mazid